

Proof Of Concept

Title: Atomic Bridge: Cross-Border Settlement Without Blind Trust

Domain: FinTech

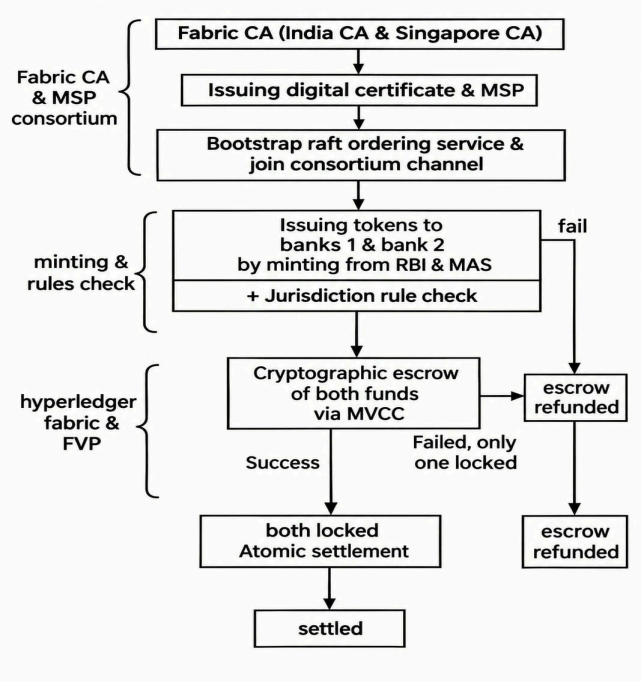
Problem Statement: Problem Statement -2

Implementing a multi-jurisdictional Hyperledger Fabric consortium network that eliminates cross-border settlement risk (Herstatt Risk) by executing multi-currency foreign exchange trades through an atomic Payment, Payment-versus-Payment (PvP) smart contract mechanism to prevent the payer loss and to minimize the timing requirements for the transaction window.

Abstract: Traditional cross-border foreign exchange settlement exposes institutions to principal default risk (Herstatt Risk) due to asynchronous, time-zone-bound clearing rails. This project presents a multi-jurisdictional Hyperledger Fabric consortium network that eliminates settlement risk by executing cross-currency FX trades via an atomic Payment-versus-Payment (PvP) smart contract mechanism. Commercial bank deposits are tokenized 1:1 against central bank reserves across distinct jurisdictional nodes. Through cryptographic escrow locking and single-state ledger transitions, counterparty assets swap simultaneously or abort entirely guaranteeing zero asset loss via automated timebound refunds if a participant defaults or disconnects mid-trade. The platform programmatically enforces local compliance rules, operating hours, and transaction limits, while employing Private Data Collections to safeguard commercial privacy and preserve regulatory oversight.







Flow For the Money swap Between Bank based in Singapore and India Jurisdiction

Features:

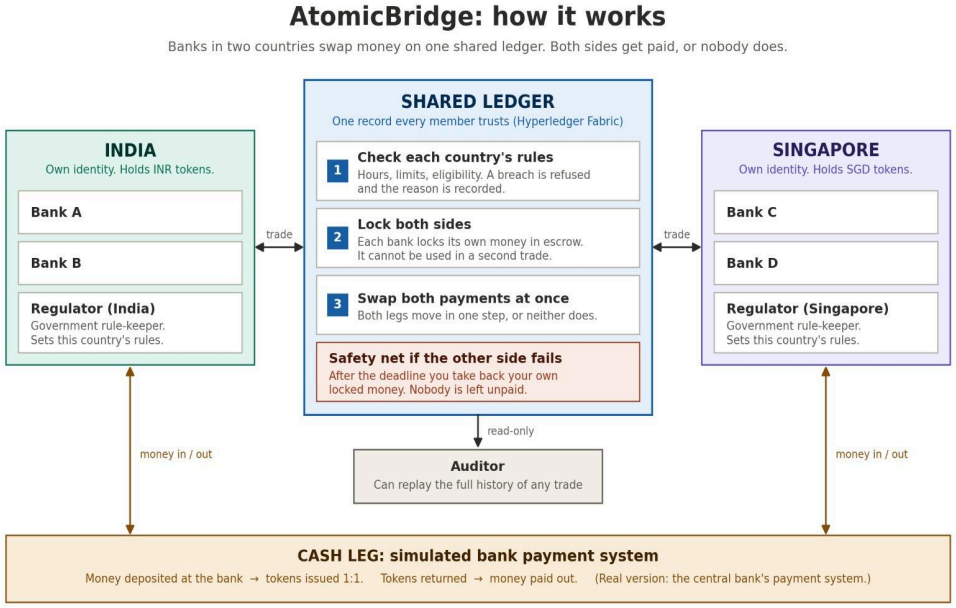
- **Real consortium:** 2 Jurisdictions, 4+ member organisations with separate CAs and MSP identities; settlement requires endorsement from both counterparties and both regulators, so none can act alone.
- **Tokens:** minted only after confirmed fund-in; held only by KYC-approved members; leave the system only through redeem (burn, then payout).
- **Atomic PvP**: INR-token leg and USD-token leg swap in one transaction on a corridor channel.
- **Jurisdictional rules:** operating hours, eligibility and limits per jurisdiction; breaches are refused and recorded on-ledger with a reason code.





- **Failure matrix:** shows what each party can determine, recover or lose when a member fails, goes silent or misses cut-off.
- **No double use:** locked funds leave the available balance; unique lock IDs plus Fabric's conflict detection block reuse.
- **Privacy and audit:** Private data collections protect trade terms and KYC data; members see only their own trades; the auditor can reconstruct any transaction's full history.
- **Cash leg:** mock RTGS API. Assumptions: 1:1 reserve at a central bank, trusted payment finality, daily reconciliation. Production path: ISO 20022 messaging.
- **Honest limits:** atomicity holds within one channel; the off-ledger payout leg carries residual risk.

Tech Stack: Hyperledger Fabric (Fabric CA, Raft ordering, private data collections), chaincode in TypeScript, Node.js/Express mock RTGS, React dashboard, Docker.



System Architecture for Atomic Bridge via Hyperledger Fabric

